any substantial reduction in entropy, which helps us stay close to the optimal location.

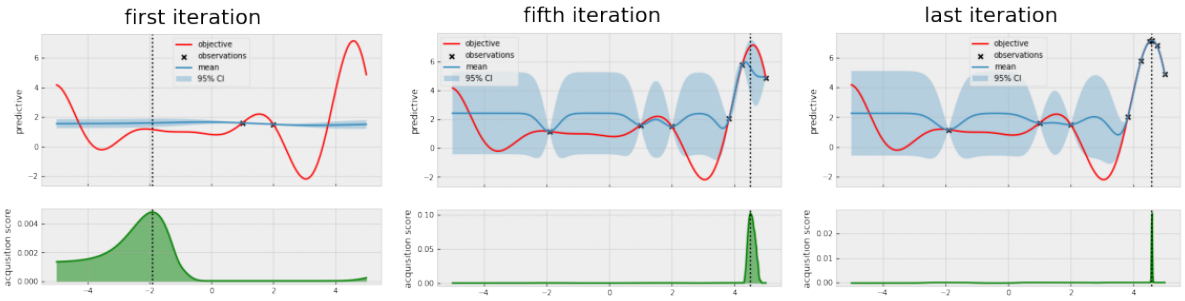


Figure 6.13 Progress made by the Max-Value Entropy Search policy. The policy quickly finds the global optimum after five queries.

We have seen that information theory gives us a principled, mathematically elegant framework for decision making that revolves around trying to learn as much as possible about a quantity of interest. This comes down to reducing the expected posterior entropy in the distribution that models the quantity we care about.

Within BayesOpt, we saw that a direct translation of this procedure poses computational challenges in modeling the location of the optimal value of the objective function, our primary search target. Instead, we shift our focus to the optimal value of the objective function itself, making the computation more tractable. Fortunately, all of this math is nicely abstracted away by BoTorch, leaving us with a convenient, modular interface that we can plug in to any optimization problem.

This is also where the second part of the book on BayesOpt policies ends. The last three chapters cover some of the most commonly used heuristics to decision making in BayesOpt and the corresponding policies, ranging from seeking improvement from the incumbent, to borrowing multi-armed bandit methods, and in this chapter leveraging information theory.

The remainder of the book takes our discussion to the next level by introducing special optimization settings that differ from what we have seen so far, where we sequentially observe a single data point at each step of the optimization. These chapters show that the methods we have learned can be translated to practical settings in the real world to accelerate optimization.

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